COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	10/14/2026	**DATE SUBMITTED:**	16/10/2026
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ACTIVITY NO. 12.1 Algorithmic Strategies

1. PROCEDURE OUTPUT

What procedures did you perform?

ILO A: Demonstrate an understanding of algorithmic strategies (Pre-Lab Review)

Table 12-1. Algorithmic Strategies and Examples

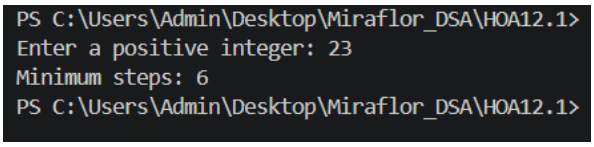
Strategy	Algorithm	Analysis
Recursion	Factorial	The function calls itself until it reaches the base case.
Brute Force	Linear Search	It checks each element one by one until it finds the target.
Backtracking	Maze Solver	it tries different paths to reach the exit. If a path is blocked, it goes back and tries another path.
Greedy	Activity Selection	It chooses the activity that finishes earliest so more activities can be selected.
Divide and Conquer	Binary Search	Divides the search area into smaller parts until the target is found.

Table 12-2. Memoization Implementation

Screenshot	

Analysis	**The program uses recursion to find the minimum steps needed to make a number become 1. It checks if subtracting 1 or dividing by 2 or 3 will give fewer steps. It also saves the answers in the memo array so it does not need to calculate them again. This makes the program faster and more efficient.**

Table 12-3. Bottom-Up Dynamic Programming Implementation

Screenshot	
Analysis	**The program uses Bottom-Up DP to find the minimum steps to change 23 into 1. It checks the possible operations and chooses the one with the fewest steps. The answer is 6 steps, and it calculates the smaller numbers first to get the final result.**

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

ILO B: Create C++ code to implement various algorithmic strategies & ILO C: Create C++ code to solve problems using learned algorithmic strategies.

Pseudocode:

```
START

Input the matrix, number of rows, number of columns,
and target cost.

Create a function countPaths(row, col, cost).

    If the current cell is outside the matrix:
        Return 0.

    Subtract the current cell value from cost.

    If the current cell is the last cell:
        If cost is 0:
            Return 1.
        Otherwise:
            Return 0.
```

C++ Code:

```
C sorting.h   main.cpp .\   main.cpp HOA12.1 ●   main.cpp ...\ILO A
HOA12.1 > main.cpp > ...
29  int main()
52  {
53      cout << "\nEnter the target cost: ";
54      cin >> cost;
55
56      int result = countPaths(matrix, 0, 0, cost, m, n);
57
58      cout << "Number of paths: " << result << endl;
59
60      return 0;
61  }
62
```

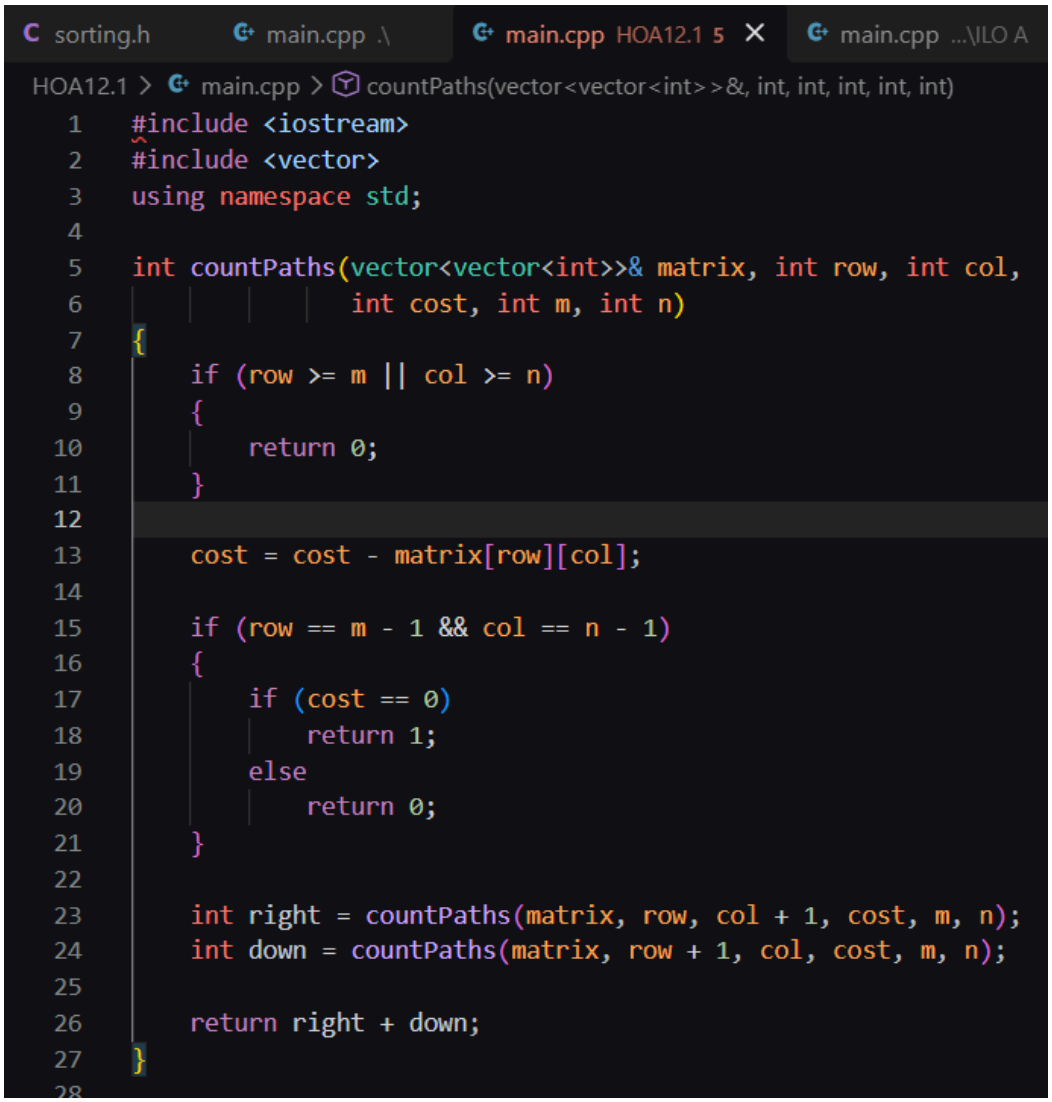
```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Admin\Desktop\Miraflor_DSA\HOA12.1> C:\Users\Admin\Desktop\Miraflor_DSA\HOA12.1\main.exe
Matrix:
4       7       1       6
6       7       3       9
3       8       1       2
7       1       7       3

Enter the target cost: 23
Number of paths: 0
PS C:\Users\Admin\Desktop\Miraflor_DSA\HOA12.1>
```

Analysis:
The program uses recursion to find the possible paths in the matrix by moving right or down. I entered 23 as the target cost, but the number of paths was 0 because no path has a total cost of exactly 23.

Screenshot of Demonstration:



sorting.h | main.cpp .\ | main.cpp HOA12.1 5 | main.cpp ...\ILO A

HOA12.1 > main.cpp > countPaths(vector<vector<int>>&, int, int, int, int, int)

```cpp

int main()
{
    int m = 4;
    int n = 4;
    int cost;

    vector<vector<int>> matrix = {
        {4, 7, 1, 6},
        {6, 7, 3, 9},
        {3, 8, 1, 2},
        {7, 1, 7, 3}
    };

    cout << "Matrix:" << endl;

    for (int i = 0; i < m; i++)
    {
        for (int j = 0; j < n; j++)
        {
            cout << matrix[i][j] << "\t";
        }
        cout << endl;
    }

    cout << "\nEnter the target cost: ";
    cin >> cost;

    int result = countPaths(matrix, 0, 0, cost, m, n);

    cout << "Number of paths: " << result << endl;
```

```cpp
    return 0;
}
```

```
Matrix:
4       7       1       6
6       7       3       9
3       8       1       2
7       1       7       3

Enter the target cost: 9
Number of paths: 0
PS C:\Users\Admin\Desktop\Miraflor_DSA\HOA12.1>
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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